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Accelerating Net Zero: CO₂ Deployment for EOR - A Comprehensive Case Study on CCUS Encompassing Reservoir Selection, Facilities Design and Operational Strategies

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The global energy industry is undergoing a transformative shift as it seeks to balance economic growth with environmental sustainability. For major energy producers like ADNOC, this transition presents both a challenge and an opportunity: maintaining production from maturing assets while reducing the carbon intensity of operations. The adoption of CO₂-EOR reflects this dual imperative, promising significant gains in oil recovery alongside meaningful contributions to carbon capture, utilization, and storage (CCUS) goals.

The pilot described herein targets a major carbonate reservoir characterized by moderate permeability contrasts between its upper and lower zones. The field, which has produced for decades under traditional waterflooding, was selected for its potential to benefit from advanced EOR techniques—specifically, the simultaneous injection of water and CO₂ (SWAG CO₂) to address bypassed oil in lower permeability intervals and improve sweep efficiency.

Introduction

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Rationale for CO₂-EOR and SWAG Implementation

Strategic Alignment with Net Zero

Company's ambition to emerge as the world's lowest-carbon-intensity oil producer has catalyzed the adoption of CO₂ injection strategies. By utilizing anthropogenic CO₂ for reservoir pressure support, ADNOC is able to simultaneously sustain production, prolong reservoir life, and achieve meaningful emissions reductions.

The SWAG scheme, in particular, was chosen for its potential to optimize oil recovery while maximizing the containment and utilization of injected CO₂.

Reservoir Suitability and Selection Criteria

The candidate reservoir was chosen based on a rigorous multi-criteria assessment, including:

- Favorable oil gravity (API), facilitating miscible or near-miscible displacement with CO₂
- Heterogeneous permeability structure, with distinct upper and lower zones
- Appropriate pressure and temperature conditions to maintain CO₂ in a supercritical state
- Substantial volumes of bypassed oil in the lower permeability zone following decades of water injection.

The reservoir's characteristics made it an ideal candidate for deploying SWAG CO₂, which promises to target unswept oil and improve overall recovery efficiency.

Overview of CO₂ Injection Schemes

Several CO₂-EOR methodologies were considered for the field, each with distinct operational philosophies and recovery mechanisms. These methodologies offers various pros and cons. Below is the comparison analysis.

Scheme	Description	Preferred Application
Continuous CO ₂ Injection	Injecting CO ₂ into a reservoir to enhance displacement efficiency is generally observed to yield higher effectiveness in homogeneous environments.	Best in the reservoirs with high miscibility and minimal heterogeneity.
Water and Alternating-Gas (WAG)	Periodic injection of water and gas (CO ₂) in alternating slugs to improve the sweep and control the gas mobility. Periodic cycle of Gas and water typically is 6 months.	Widely used in heterogeneous reservoirs to reduce the CO ₂ channeling and to improve the volumetric sweep efficiency.
Simultaneous Water and Gas (SWAG)	Concurrent injection of water and CO ₂ , often through the separate wellbores or commingled streams, targeting different reservoir intervals.	This approach enhances the vertical and areal sweep. It is mostly suitable for reservoirs with significant permeability contrast.
Huff and Puff	CO ₂ is injected into a well, left to soak, then produced from the same well. Repeating these Huff and Puff cycles enhances recovery.	Most common in the tight or fractured reservoirs, or mature fields with limited injectivity.

The SWAG scheme was ultimately selected for the pilot, offering a hybrid solution that leverages the strengths of both WAG and continuous injection by simultaneously targeting distinct reservoir intervals.



Figure 1—Typical CO₂ Injection Scheme

Design and Operational Philosophy

Well Configuration and Placement

The SWAG CO₂ pilot is structured around a cluster of wells, strategically positioned to ensure the optimal fluid diversion and monitoring:

- Oil production wells, targeting, the most favorable drainage area.
- Horizontal water injector, placed in the upper reservoir interval
- Horizontal CO₂ injector, focused on the tighter lower interval
- Observer wells, arrayed both up-dip and down-dip to provide comprehensive saturation profiling and displacement front monitoring

Spacing between the injector and producer wells is maintained, striking a balance between effective pressure communication and sweep coverage.

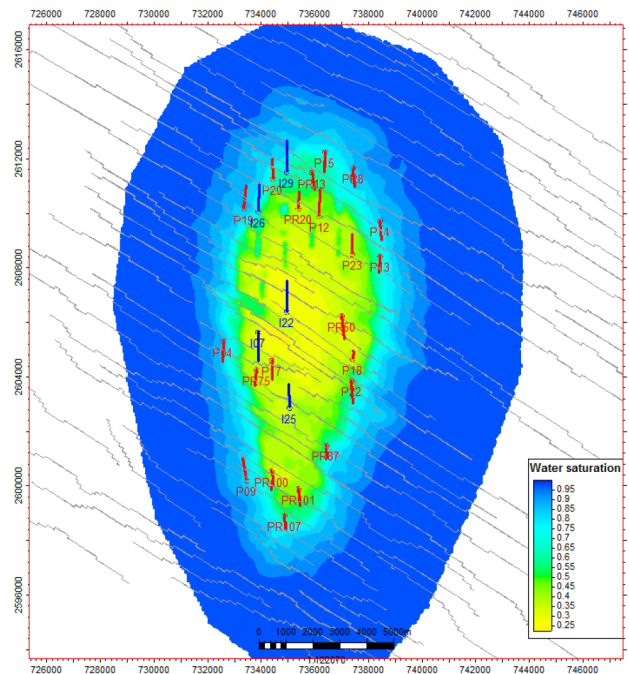


Figure 2—Well Placement Pattern

Surface Facilities and CO₂ Handling

A dedicated CO₂ sourcing and transportation network was established to deliver CO₂ in a supercritical state from emission sources to the injection field. Key design elements include:

- More than 45 Km of Carbon Steel, 6 mm Corrosion allowance Pipelines, ASME 2500# rated for high-pressure, high-density CO₂ transport to the site.
- Sectionalization of the network introduced for operational flexibility and leak management
- Real time leak detection systems based on molecular composition analysis
- Downhole asphaltene inhibitor injection skids to manage production risks.
- De emulsifier injection skid.
- Mobile methanol and flare system.

Commissioning of the whole project was executed in phased Ready-for-Commission (RFC) stages to minimize start-up risk and ensure smooth integration with existing field infrastructure.

- RFC – 1, commissioning of water injection wells.

- RFC -2, commissioning of Oil producing wells on Naturally flowing mode.
- RFC – 3, commissioning of utilities and chemical skids.
- RFC – 4, commissioning of CO₂ pipeline network.

Operational Parameters and Safety

Operating pressures are maintained at 190 bar, which is above the supercritical threshold for CO₂ to ensure maximum injectivity and containment. Extensive safety protocols and real-time monitoring technologies—including permanent downhole gauges and distributed temperature sensing—enable prompt detection of anomalies and rapid response to operational issues.

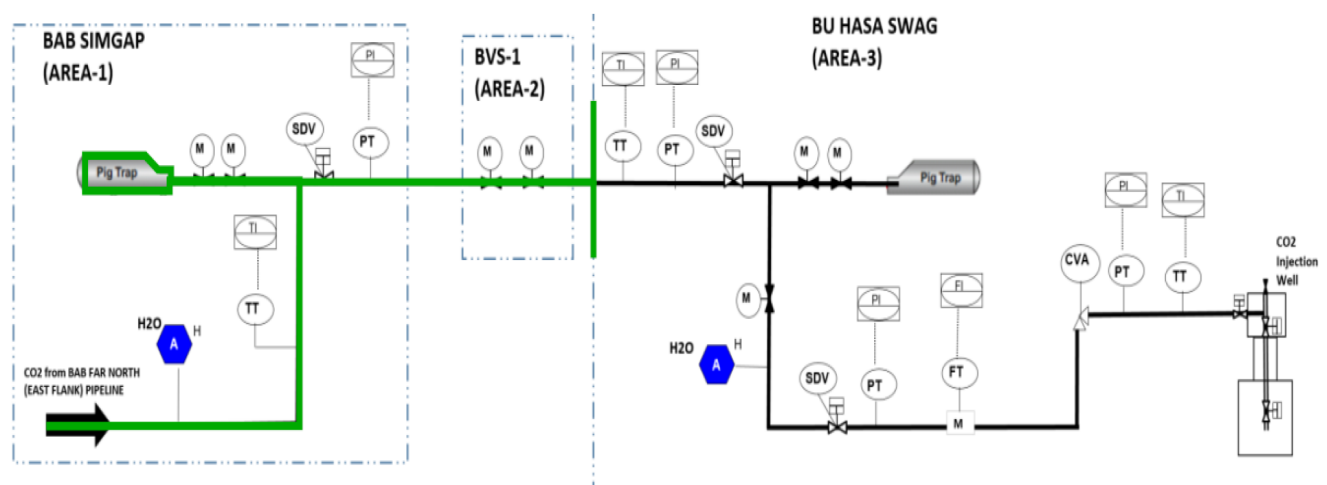


Figure 3—Typical schematic of CO₂ pipe line network

Sectionalization / Inventory Management

Due to vast inventory for CO₂ in the Pipeline, Pipeline sectionalization was considered and achieved by providing Block Valve Station in the mid way of 45 km Pipeline. Block Valve Station is equipped with high pressure CO₂ vent stack and Nitrogen cylinder package and is linked with a leak detection system using mass balance technology with negative pressure wave.

Provision of Block valves station on CO₂ Pipeline provides the following advantages:

- Reduces the volume of released CO₂ gas in case of Pipeline containment failure/leakage
- Facilitate operation and maintenance by reducing the amount of CO₂ that needs to be handled
- Facilitate in reducing the requirement of Nitrogen for charging the header after the cold startup.
- Risk reduction.

High-pressure CO₂ vent stack is used for the safe venting at the Pipeline's block valve station. High pressure venting is essential for ensuring no risks of CO₂ ice formation inside the Pipeline and the vent piping nor the risk of significant temperature drop inside the system while depressurizing. These high pressure CO₂ vent stacks are provided with remotely operated valve (MOV) and pressure, temperature transmitters downstream of MOV, which facilitates the remote depressurization of system.

Block valve station venting configuration shall be designed with redundancy of depressurizing the Pipeline from both sides of the BVS. A 4" bypass connection is provided across the BVS for this purpose.

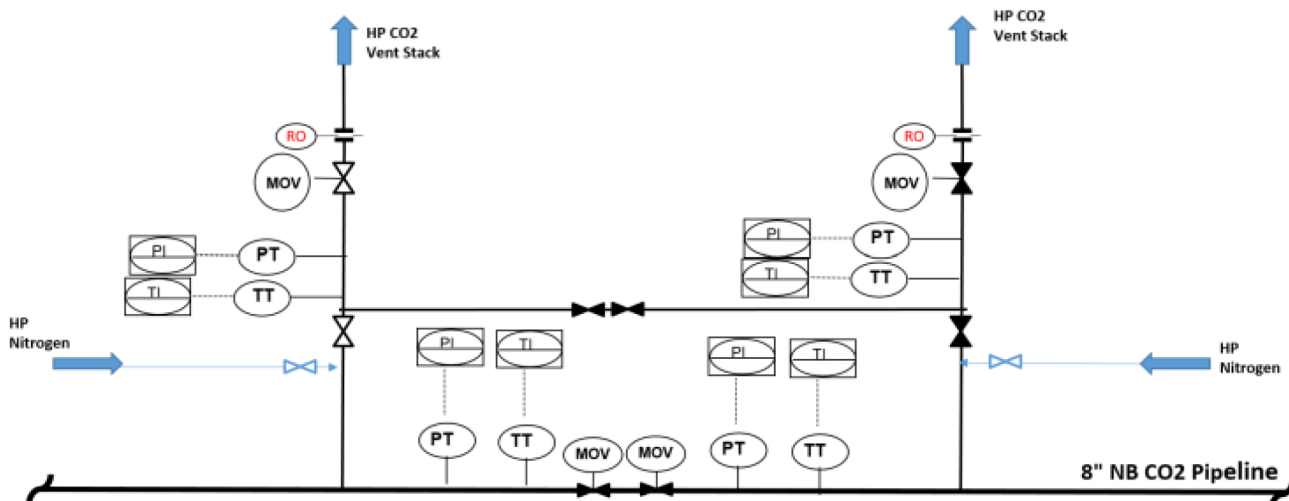


Figure 4—Block Valve station arrangement

Chemical System

This section describes the various chemical systems implemented to ensure the safety and integrity of the CO₂ SWAG Pilot,

Methanol System. Methanol is considered for injection in the network before CO₂ depressurization event from the min transfer line or individual well flow lines. A mobile methanol package is therefore, provided to be utilized whenever and wherever is required. The mobile Methanol injection package design is implemented with the below safety technical requirements:

- Chemical storage tanks/vessel is designed to a design pressure of 3.5 barg & Full Vacuum
- All safety features, as deemed necessary are implemented for safe operation, maintenance of mobile methanol chemical injection packages.
- Vessel is fitted with pressure safety valve. Pressure Safety Valve sized for both fire as well as blanketing PRV failure
- Storage vessel is top sealed and not open to atmosphere during offline condition
- Methanol storage is provided with nitrogen blanketing connection and is used during package operation (during pump out) as well as during filling.

Nitrogen System. Portable Nitrogen rack system with all the associated necessary accessories, inclusive of regulators, valves, instruments, manifold, steel rack and junction boxes, mobile features, quick coupling connections etc., for safe operation and handling

Mobile N₂ cylinder rack/skid design is provided with movable and portable features to facilitate the mobilization and re-use of the skid at different locations, with Quick coupling connections. Nitrogen rack is design in four arrays, which enables removal and charging of cylinders without impacting the availability of whole system.

Mobile Flare Package. Mobile flare skid is provided for depressurization of gas lift lines during gas lift line shutdown or any maintenance activity. The design pressure is to be considered as 8.6 barg whereas the mobile flare metallurgy is finalized as Stainless Steel with MDMT of -105 °C and the same was demonstrated through MDMT depressurization analysis.

Mobile flare is truck mounted with retractable stack, design is in compliance with dispersion studies.

De-emulsifier and Asphaltene Inhibitor Package. Chemical dosing skids are provided for as follows:

- Dedicated demulsifier Package to be utilized for the new test separator as the wells will be on continuous testing to observe any CO2 breakthrough.
- A Wellhead Downhole Asphaltene Injection Package is also provided to inject asphaltene up to 500 ppm.

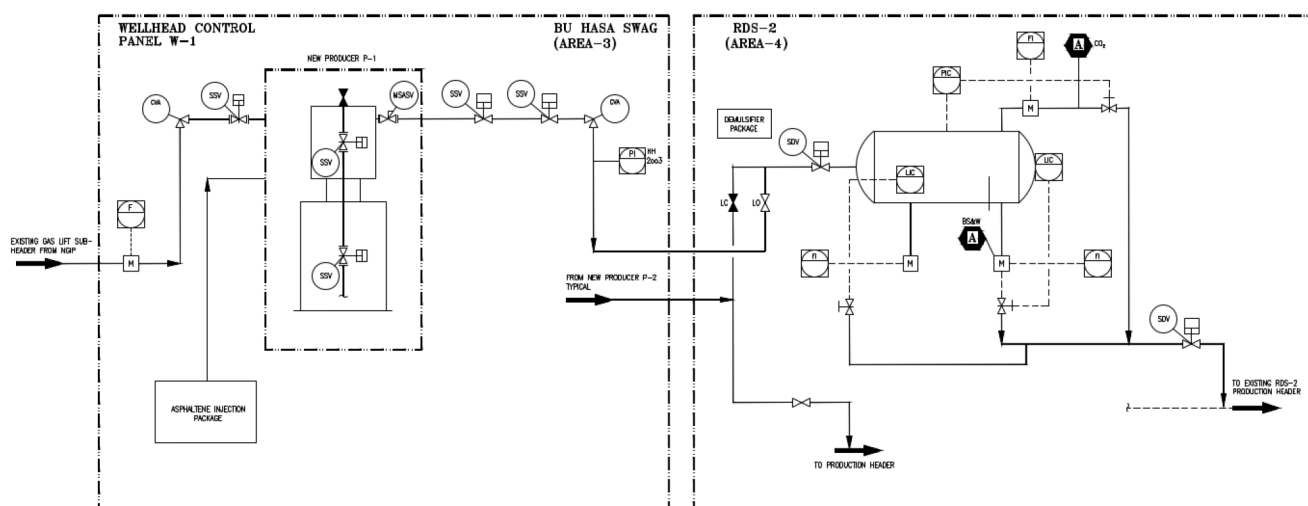


Figure 5—Chemical Injection skids

Reservoir Management and Surveillance

Saturation Monitoring and Data Acquisition

The observer wells were equipped with state-of-the-art logging and surveillance tools to track the movement of injected water and CO₂. Objectives of this arrangement includes:

- Mapping the advance of displacement fronts in both upper and lower intervals.
- Quantifying the sweep efficiency, breakthrough timing, and residual oil saturation.
- Assessing the effectiveness of vertical conformance between injection zones

Real-time Data Integration

A digital field platform integrates data streams from wellhead sensors, downhole equipment, and surface facilities. Best use of available technologies were made to effectively evaluate condition of surface facilities and also for the proactive reservoir management, informing operational adjustments and supporting the de-risking of full-field SWAG implementation.

All surface parameters including CO2 line pressure, flow rates, Wellhead parameters and functionalities are monitored and routed to the central control room. All wells are equipped with remote monitoring and control applicability to minimize human interference and part of lean operating mode.

Technical Uncertainties and Risk Mitigation

Identification of Key Risks

A comprehensive dynamic risk assessment was conducted prior the execution of pilot with special focus on the:

- CO₂ containment and potential for leakage or migration outside the target zones
- Uncertain of injectivity in the lower reservoir interval
- Changes in the reservoir chemistry, behavior or wettability affecting the CO₂ mobility
- Potential asphaltene precipitation and scaling risks in the Oil producer wells.
- Operational upsets due to equipment failure or unforeseen reservoir responses over the time.

Derisking strategy and technical uncertainties

In order to de-risk the full field Development in the selected candidate reservoir, We carried out holistic review to identify the main risks and uncertainties of the process in totality. Several on paper and actual exercises were conducted to de-risk and address all potential uncertainties.

Table 1—SWAG Technical Uncertainties.

No.	Potential Risk Identified	Detail	MITIGATION
1	Sub surface Faults and lateral communication	Early Break through and Loss of early containment	Tried to avoid faulted areas during location selection
2	History match quality	Can affect pilot prediction and also can result in additional uncertainties on pilot design.	Evaluated the impact on pilot results i.e. uncertainty analysis and improved history match
3	CO2 sweep efficiency	Loss of CO2 confinement in targeted layers/pilot area leaving behind upswept oil	Sensitivities analysis were performed on design & geological parameters (permeability layer contrasts, High permeability streaks, heterogeneities, Kv/Kh)
4	CO2 displacement efficiency	Not able to reduce the oil saturation because of ineffective displacement	Reviewed & sensitized SCAL/PVT parameters to assess impact of EOR results
5	CO2 miscibility at reservoir condition	Loss of miscibility	CO2 MMP verification & ensure Injection and reservoir pressure remain above MMP.
8	SCAL Uncertainties	Uncertainty can be found in the Relative Permeability curves used in the dynamic model	Relative permeability sensitivity was done within the range of uncertainty, while honoring the endpoints, to assess the impact of Kr on fluid mobilities.
9	High Permeability Streaks	Based on high interval frequency RCA and mini-perm experiment conducted, vuggy intervals were found	According to the findings, sensitivity analysis was conducted sensitizing on how transmissible these intervals are and their impact on pilot
9	CO2 supply and CO2 purity	Must maintain CO2 specs to have miscibility & avoid flow assurance issues. Maintain CO2 supply at required rates	Keep track of CO2 specs and ensure ON Spec CO2 delivery in collaboration with supplier. Adjust production rates to avoid depletion in pilot area in case of CO2 shortage.
10	CO2 pipe leaking	Leak of CO2 at high Pressure with H2S content	Proper metallurgy, leak detection, depressurization system to be in place & minimize pipe length.
11	Asphaltene Deposition	Asphaltene deposits in the reservoir, wellbore and facilities due to CO2 injection	For reservoir: maintain pressure > Asphaltene precipitation level For wells and surface facilities: plan for asphaltene inhibitor package
12	Construction in Congested Area	Construction related risk	Route selection was carried out in consultation with master plan. Various Constructability reviews were performed to ensure safe execution.
13	Consistent quality of water injection	Risk of change of water quality due to change in source of water from Aquifer to Sea water.	Communication established with various stakeholders to ensure that quality of water / source remained the consistent.

Having completed the initial qualitative assessment and listing all of the uncertainties and risks. Subsurface uncertainties were assessed on both cumulative oil productions in addition to recovery factor through numerical simulation.

The main outcomes indicated that uncertainties in Kv/Kh and High permeability streaks could be the most detrimental to oil recovery, while CO₂ purity could be relaxed down to approximately 95% with no major impact. CO₂ injection rates gave best recovery at 6.0 MMscfd with simultaneous water injection rates in the upper intervals at 5000 bbls/d.

Mitigation Strategies

Risks were addressed through a combination of design features, real-time surveillance, and contingency planning:

- Observer wells for early detection of CO₂/water breakthrough
- Sectionalized surface facilities for rapid isolation and management of leaks
- Continuous chemical injection and scale management programs
- Flexible operational protocols to accommodate dynamic reservoir responses

Pilot Performance and Results

Recovery Factor and Reservoir Response

The pilot has delivered promising results, achieving a recovery factor of up to 54% with a Hydrocarbon Pore Volume Injected (HCPVI) of 1.67. Key performance indicators include:

- Effective containment and sweep in lower permeability zones
- Noticeable increase in oil production from previously bypassed intervals
- Minimal CO₂ breakthrough in production wells, confirming vertical conformance
- Stable reservoir pressures and absence of major operational upsets

Lessons Learned and Scalability

The SWAG pilot has demonstrated the technical and economic feasibility of CO₂-EOR in ADNOC Onshore's mature carbonate fields. Lessons learned include:

- The critical importance of well placement and spacing in maximizing sweep
- The value of real-time surveillance for managing technical uncertainties
- The necessity of robust surface infrastructure to support CO₂ handling and monitoring

These insights will inform the design of future pilots and the eventual scale-up to full-field implementation.

Integration with Sustainability and CCUS Objectives

CO₂ Utilization and Climate Benefits

By utilizing anthropogenic CO₂ captured from industrial sources, the pilot advances both hydrocarbon recovery and carbon management goals:

- Permanent sequestration of significant CO₂ volumes in the subsurface
- Reduction of greenhouse gas emissions relative to conventional production
- Contribution to ADNOC's net-zero targets and national CCUS commitments.

Broader Industry Implications

The successful deployment of SWAG CO₂ at ADNOC Onshore sets a benchmark for other energy producers in the region and globally. The project demonstrates how innovation and operational excellence can reconcile energy security with climate action, providing a replicable model for sustainable oil production.

Future Directions and Recommendations

Building on the achievements of the pilot, several avenues of further research and deployment are recommended:

- Extension of SWAG CO₂ schemes to other fields with suitable reservoir characteristics
- Integration with renewable energy sources and green hydrogen for holistic decarbonization
- Continued investment in digital surveillance and advanced reservoir modeling
- Collaboration with technology providers, research institutions, and regulatory bodies to refine best practices

Conclusion

ADNOC Onshore SWAG CO₂ pilot stands as a landmark achievement in the journey toward sustainable hydrocarbon production. By combining state of the art Enhanced Oil Recovery (EOR) methods with comprehensive carbon management strategies, ADNOC is at the forefront of the global energy transition. The project not only advances operational excellence and innovation but also presents a blueprint for balancing energy security with environmental responsibility. Lessons learned from this pilot will lay strong foundation for future projects and developments, supporting ADNOC's net-zero ambitions and setting a precedent for the industry. Ultimately, this initiative demonstrates that climate action and efficient resource development can go hand in hand.

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Appendix C: Glossary

CO ₂ -EOR	: Carbon Dioxide Enhanced Oil Recovery
SWAG	: Simultaneous Water and Gas injection
WAG	: Water-Alternating-Gas injection
HCPVI	: Hydrocarbon Pore Volume Injected
CCUS	: Carbon Capture, Utilization, and Storage
RFC	: Ready for Commission
BVS	: Block Valve station.

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